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Interview Preparation Series

On-Premises to Azure Migration

Migration Types · Use Cases · Scenario-Based Q&A · FAQs

A complete, interview-ready reference for cloud & database migration roles

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1. How to Use This Guide

This document is structured the way a **real migration interview flows** — from strategy fundamentals, to the tooling, to hands-on scenarios and rapid-fire FAQs. It covers **both infrastructure and database migration**, so it fits general Azure migration roles as well as data-platform-specific ones.

- **Sections 2–4** build the framework: strategies (the 6 R's), workload types, and the Azure tooling ecosystem.
- **Section 5** is a database-migration deep dive — SQL Server to Azure SQL DB / Managed Instance / VM, online vs offline methods.
- **Section 6** walks through end-to-end use cases you can narrate as war stories.
- **Sections 7–8** are scenario-based Q&A and FAQs — the parts an interviewer actually drills on.

TIP — Interview framing

When asked "how would you migrate X", always answer in a fixed order: (1) Assess & discover, (2) choose a strategy/target, (3) pick a migration method (online vs offline), (4) cutover & validate, (5) optimise & decommission. Interviewers reward a repeatable framework over a list of tools.

2. Migration Strategies — The 6 R's

Every migration decision maps to one of Microsoft's (and Gartner's) migration strategies. Interviewers expect you to name them, define them, and — critically — say **when** you'd pick each one.

Strategy	What it means	Effort / Risk	Typical Azure target
Rehost (Lift & Shift)	Move the workload as-is, no code changes.	Low effort, low risk, least optimisation	Azure VM (IaaS), SQL Server on Azure VM
Replatform (Lift, tinker & shift)	Minor optimisations without changing core architecture.	Low–medium	Azure SQL Managed Instance, App Service, Azure DB for PostgreSQL/MySQL
Refactor / Repackage	Containerise or make small code changes to fit PaaS.	Medium	AKS, Azure Container Apps, App Service
Rearchitect	Redesign the app, often monolith → microservices.	High effort, higher payoff	AKS, Functions, Cosmos DB, Event Hubs
Rebuild	Rewrite from scratch as cloud-native.	Highest effort	Serverless / PaaS-native stack
Replace (Repurchase)	Retire the app and adopt a SaaS product.	Varies	Microsoft 365, Dynamics 365, 3rd-party SaaS

INFO — Two more R's you should mention

Retain (revisit) — keep certain workloads on-prem for now (latency, compliance, dependency) and run hybrid.

Retire — decommission workloads no longer needed; discovered during assessment and often 10–20% of an estate.

2.1 How to choose a strategy

- **Speed / deadline-driven (data-centre exit)** → Rehost first, optimise later. Fastest path off-prem.
- **Want to cut OS/patching/HA overhead** → Replatform to PaaS (e.g. SQL MI removes OS + backup + patching burden).
- **App is a scaling / agility bottleneck** → Rearchitect/Rebuild for elasticity.
- **Commodity capability (email, CRM)** → Replace with SaaS.

3. Types of On-Prem → Azure Migration (by Workload)

Migration is rarely one thing — a real estate mixes several workload types. Know the tool for each.

Workload type	What moves	Primary Azure tool	Target
Servers / VMs	Physical + VMware/Hyper-V VMs	Azure Migrate: Server Migration	Azure VMs
Databases (SQL)	SQL Server, Oracle, MySQL, PostgreSQL	Azure Database Migration Service (DMS), DMA, SSMA	Azure SQL DB / MI / VM, Flexible Servers
Web apps	.NET / Java web apps + IIS sites	App Service Migration Assistant	Azure App Service
Files / storage	File shares, unstructured data	Storage Migration Service, Azure File Sync, AzCopy, Data Box	Azure Files, Blob Storage
Virtual desktops	On-prem VDI / RDS	Azure Virtual Desktop	AVD host pools
Identity	On-prem Active Directory	Microsoft Entra Connect (AD Connect)	Microsoft Entra ID
Big data / analytics	Hadoop, data warehouses	Azure Data Factory, Synapse	Synapse, Fabric, ADLS

3.1 Migration by hosting model (IaaS → PaaS → SaaS)

- **IaaS** — most control, most responsibility. You still patch the OS and manage HA. Best for lift-and-shift and unsupported/legacy workloads.
- **PaaS** — Azure manages OS, patching, HA and backups. Less control, far lower operational overhead. The strategic target for most modernisation.
- **SaaS** — consume the whole application; you manage only configuration and data.

4. The Azure Migration Toolbox

Azure Migrate is the central hub — a single portal for discovery, assessment and migration across servers, databases, web apps and VDI. Around it sit specialised tools.

4.1 Azure Migrate — the hub

- **Discovery & assessment** — inventories servers, maps dependencies, and produces right-sizing + cost (TCO) estimates using an on-prem appliance.
- **Server Migration** — replicates and migrates VMs. Agentless (VMware/Hyper-V) or agent-based.
- **Dependency analysis** — agentless (via appliance) or agent-based (needs MMA + Dependency agent) to group servers into migration waves.

4.2 Database & data movement tools

Tool	Purpose	When to use
Data Migration Assistant (DMA)	Assess compatibility issues & feature parity, then migrate schema+data.	Pre-migration assessment for SQL → Azure SQL
Azure Database Migration Service (DMS)	Managed service for at-scale online/offline DB migrations.	SQL → Azure SQL DB / MI, minimal-downtime moves
SQL Server Migration Assistant (SSMA)	Heterogeneous migration (Oracle/MySQL/DB2/Access → SQL).	Cross-engine migrations
Azure Data Studio + extensions	Azure SQL Migration extension drives assessment + DMS.	Modern GUI-driven SQL migrations
Data Box (Disk / Box / Heavy)	Offline bulk transfer via physical appliance shipped to you.	TB–PB of data, poor bandwidth
AzCopy / Storage Migration Service	Copy files & blobs; migrate Windows file servers.	Unstructured data, file shares

TIP — DMA vs DMS — a classic interview trap

DMA (Data Migration Assistant) is the desktop assessment/small-migration tool — it finds blocking issues and feature-parity gaps.

DMS (Database Migration Service) is the managed Azure service that performs the actual scaled migration, including online (near-zero-downtime) mode. You assess with DMA, then migrate with DMS.

5. Database Migration Deep Dive (SQL Server → Azure)

This is where a DBA earns the offer. Two decisions drive everything: (1) which target, and (2) online vs offline.

5.1 Choosing the target

	Azure SQL Database	SQL Managed Instance	SQL Server on Azure VM
Model	PaaS	PaaS	IaaS
Compatibility	Database-scoped; some T-SQL gaps	~Near 100% instance-level	100% (full SQL Server)
Best for	New / cloud-native apps, single DBs	Lift-and-shift of whole instances	Legacy, OS access, unsupported features
SQL Agent	No (use Elastic Jobs)	Yes	Yes
Cross-DB queries	Limited (elastic query)	Yes	Yes
You patch OS?	No	No	Yes
HA built-in	Yes (99.99%+)	Yes	You configure (AG/FCI)

INFO — Rule of thumb

Need the whole instance (SQL Agent jobs, cross-DB, CLR, Service Broker) with least code change → Managed Instance.

Single modern app database, want fully managed & serverless options → Azure SQL Database.

Need OS-level access, third-party agents, or an unsupported feature → SQL Server on Azure VM.

5.2 Migration methods — Offline vs Online

Offline = application is down for the duration of the data copy. **Online** = source stays live; changes are continuously synced and you cut over with seconds–minutes of downtime.

Method	Mode	Target	Notes
Backup & restore (to URL / VM)	Offline	Azure VM	Simplest for IaaS; native .bak restore
BACPAC export/import	Offline	SQL DB / MI	Schema+data; not for very large DBs
DMS — offline	Offline	SQL DB / MI	Managed, scaled one-time migration
DMS — online	Online	SQL DB / MI	Continuous sync, minimal downtime
Managed Instance link	Online	SQL MI	Log-based near-real-time replica; fast cutover
Log shipping	Online-ish	VM / MI	Restore log chain, then final tail-log cutover
Transactional replication	Online	SQL DB / MI / VM	Granular; good for phased cutovers
Native backup to URL	Offline	VM	Backup to Blob, restore in Azure

5.3 Example — minimal-downtime SQL MI migration with DMS (online)

A typical narrative you can walk an interviewer through:

1. Run DMA / Azure SQL Migration assessment to catch blocking compatibility issues.
2. Provision target Azure SQL Managed Instance sized from the assessment.
3. Set up network line-of-sight (ExpressRoute / VPN / private endpoint) and an SMB share or Blob container for backups.
4. Start DMS online migration — it restores full + log backups continuously to keep the target in sync.
5. Validate data, objects, logins and jobs on the target.
6. Perform cutover during a low-traffic window: stop writes, let final logs apply, repoint the app connection string.
7. Monitor, then decommission the source after a bake-in period.

Illustrative pre-migration check with dbatools before cutover:

```
# Compare logins, jobs and DB objects between source and target
$src = 'ONPREM\PROD'
$tgt = 'sqlmi-champs.<zone>.database.windows.net'

# Migrate logins (SIDs preserved) so app auth doesn't break at cutover
Copy-DbLogin -Source $src -Destination $tgt -ExcludeSystemLogins

# Copy SQL Agent jobs (MI supports Agent; SQL DB does not)
Copy-DbAgentJob -Source $src -Destination $tgt

# Validate row counts on critical tables post-sync
Invoke-DbQuery -SqlInstance $tgt -Database Sales -Query `
'SELECT COUNT(*) AS Orders FROM dbo.Orders;'
```

WATCH OUT — Cutover gotchas interviewers love

Orphaned users — logins migrated without matching SIDs break app authentication. Migrate logins with SIDs preserved.

SQL Agent jobs, linked servers, and credentials do NOT move with the data — script them separately.

Don't forget DNS/connection-string change and TLS/firewall rules, or the app can't reach the new instance.

6. End-to-End Use Cases

Concrete, narratable scenarios — the kind of "tell me about a migration you did" answers that land well.

Use Case 1 — Data-centre exit under a deadline

Situation: Lease expiring in 4 months; 120 mixed Windows/Linux VMs and 30 SQL databases.

Approach: Rehost the VMs with Azure Migrate: Server Migration; lift SQL to SQL Server on Azure VM (fastest, full compat). Optimise to PaaS later.

Why: Deadline beats optimisation. Move first, modernise in place afterwards.

Use Case 2 — Reduce DBA operational overhead

Situation: A small team drowning in patching, backups and HA maintenance for 15 SQL instances.

Approach: Replatform to Azure SQL Managed Instance via DMS online migration. Near-100% compatibility means minimal app change.

Outcome: Automatic backups, patching and built-in HA (99.99%) — operational toil drops sharply.

Use Case 3 — Modernise a monolithic web app

Situation: A .NET monolith on IIS that can't scale for seasonal peaks.

Approach: Refactor/Rearchitect — App Service for the web tier, Azure SQL Database for data, and split hotspots into Functions. Autoscale handles peaks.

Use Case 4 — Huge dataset, poor bandwidth

Situation: 80 TB of archive + a 6 TB database to move; site has a thin WAN link.

Approach: Ship data offline with Azure Data Box for the archive; use backup-to-URL / DMS for the DB. Avoid saturating the WAN.

Use Case 5 — Hybrid, phased migration

Situation: Compliance requires some data stays on-prem for now.

Approach: Retain sensitive workloads on-prem; connect via ExpressRoute/VPN; use Azure Arc to manage on-prem servers from Azure. Migrate the rest in waves.

7. Scenario-Based Interview Questions

These test judgement, not memory. Answer with the framework: assess → choose target → choose method → cut over → validate.

S1 You must migrate a 4 TB SQL Server database to Azure with less than 15 minutes of downtime. How?

A: Use an online method so the source stays live during the bulk copy. For Managed Instance, the Managed Instance link or DMS online migration both stream log backups continuously; cut over by applying the final log and repointing the app.

A: Steps: assess with DMA → provision target sized from the assessment → establish network line-of-sight → start continuous sync → validate objects/logins/jobs → cut over in a low-traffic window → monitor and decommission.

- Avoid BACPAC or plain backup/restore here — they're offline and can't hit a 15-minute window at 4 TB.

S2 A legacy app uses cross-database queries, CLR and SQL Agent jobs. Which Azure SQL target?

A: Azure SQL Managed Instance. It supports instance-scoped features — cross-DB queries, SQL Agent, CLR, Service Broker, linked servers — so the app needs minimal change. Azure SQL Database would break the cross-DB and Agent dependencies; a VM works but keeps the OS/patching burden.

S3 Assessment flags 200 blocking compatibility issues on a DB destined for Azure SQL Database. What now?

A: Three options: (1) remediate the issues (rewrite unsupported T-SQL/features) if the app truly needs PaaS SQL DB; (2) re-target to Managed Instance, which resolves most instance-level gaps with far less rework; (3) if features are genuinely unsupported anywhere in PaaS, target SQL Server on Azure VM.

A: The right call is usually re-target to MI unless there's a strong reason for SQL DB.

S4 The customer wants to exit their data centre in 8 weeks. 300 servers. What's your plan?

A: Prioritise speed: Rehost with Azure Migrate: Server Migration. Discover + assess with the Azure Migrate appliance, run dependency analysis to build migration waves, pilot a low-risk wave, then migrate in batches with test-migrations before each cutover. Modernise to PaaS after the estate is safely in Azure.

S5 How do you migrate 100 TB of file-server data with limited WAN bandwidth?

A: Offline transfer with Azure Data Box (Box or Heavy for this volume). Copy data onto the appliance on-site, ship it back, and Microsoft loads it into your storage account. Use Storage Migration Service / Azure File Sync for the residual delta and cutover.

S6 After cutover, the application can't authenticate to the new SQL instance. Likely cause?

A: Orphaned users — logins were migrated but their SIDs don't match the database users, or the logins weren't migrated at all. Fix by recreating logins with the original SIDs (or run `sp_change_users_login / ALTER USER ... WITH LOGIN`). Also verify firewall rules, connection string, and TLS settings.

```
-- Detect orphaned users in the migrated database
SELECT dp.name AS orphaned_user, dp.type_desc
FROM sys.database_principals dp
LEFT JOIN sys.server_principals sp ON dp.sid = sp.sid
WHERE sp.sid IS NULL
      AND dp.type IN ('S','U','G')
      AND dp.name NOT IN ('dbo','guest','sys','INFORMATION_SCHEMA');
```

S7 How do you validate a database migration was successful before decommissioning the source?

A: Compare object counts and row counts on critical tables, verify logins/users/permissions, confirm SQL Agent jobs and linked servers exist, check that DBCC CHECKDB is clean on the target, run application smoke tests, and confirm performance is acceptable (compare Query Store / execution plans). Keep the source available read-only for a bake-in period as a rollback path.

S8 Cost came in higher than expected after migration. How do you optimise?

A: Right-size VMs/DBs from actual utilisation, use Azure Hybrid Benefit for existing SQL/Windows licences, apply Reserved Instances / savings plans for steady workloads, use serverless/auto-pause tiers for spiky dev/test databases, and clean up orphaned disks/IPs. Azure Advisor and Cost Management surface the recommendations.

8. Rapid-Fire FAQs

Q. What is the difference between lift-and-shift and replatforming?

Lift-and-shift (Rehost) moves a workload unchanged to IaaS. Replatforming makes small optimisations — e.g. moving a SQL database to Managed Instance — without redesigning the application.

Q. What is Azure Migrate?

A central hub in the Azure portal for discovering, assessing and migrating servers, databases, web apps and virtual desktops from on-prem (or other clouds) to Azure. It provides an on-prem appliance for agentless discovery, dependency mapping, right-sizing and TCO estimates.

Q. DMA vs DMS vs SSMA — what's the difference?

DMA (Data Migration Assistant) assesses compatibility and does small migrations. DMS (Database Migration Service) is the managed Azure service for large-scale online/offline migrations. SSMA (SQL Server Migration Assistant) handles heterogeneous migrations from Oracle, MySQL, DB2 or Access into SQL.

Q. When would you choose SQL Server on Azure VM over a PaaS option?

When you need OS-level access, third-party agents, unsupported SQL features, specific SQL versions, or 100% compatibility with minimal change — and you're willing to keep managing patching, backups and HA yourself.

Q. What is the difference between online and offline migration?

Offline migration takes the source down for the whole data copy. Online migration keeps the source live and continuously syncs changes, so downtime is limited to a short final cutover — used when the business can't tolerate a long outage.

Q. What is Azure Hybrid Benefit?

A licensing benefit that lets you apply existing on-prem Windows Server and SQL Server licences with Software Assurance to Azure, significantly reducing compute/licensing cost after migration.

Q. How do you connect on-prem to Azure during migration?

Via Site-to-Site VPN for smaller/temporary connectivity, or ExpressRoute for dedicated, private, high-bandwidth links. Private endpoints keep PaaS traffic off the public internet.

Q. What is Azure Arc's role in migration?

Azure Arc extends Azure management (policy, monitoring, security, SQL management) to on-prem and multi-cloud servers. It's useful in hybrid/phased migrations to govern workloads that stay on-prem before or during their move.

Q. What are the phases of the Cloud Adoption Framework (CAF) for migration?

Strategy → Plan → Ready (landing zone) → Adopt (Migrate / Innovate) → Govern → Manage. The migration itself follows Assess → Migrate → Optimise → Secure & Manage.

Q. What is a landing zone?

A pre-configured, governed Azure environment (networking, identity, security, management, policy) set up before workloads land — so migrated resources are compliant and secure from day one.

Q. How do you handle a database too large for BACPAC?

Use DMS (online or offline), backup/restore to a VM, or the Managed Instance link for MI. BACPAC is practical only for small-to-medium databases.

Q. What's the difference between agentless and agent-based discovery in Azure Migrate?

Agentless uses the Azure Migrate appliance to query VMware/Hyper-V hosts with no software on each VM. Agent-based installs the MMA and Dependency agent on servers for deeper dependency data — needed for physical servers and detailed app-level mapping.

Q. What is Azure Data Box used for?

Offline, bulk data transfer. Microsoft ships a physical appliance; you copy TB–PB of data locally and return it, avoiding slow or costly network transfer over the WAN.

Q. After migration, how do you reduce cost?

Right-size resources from real utilisation, apply Azure Hybrid Benefit, buy Reserved Instances/savings plans for steady workloads, use serverless/auto-pause for spiky databases, and remove orphaned disks/IPs. Use Azure Cost Management and Advisor.

Q. What is a migration wave?

A grouped batch of interdependent servers/apps migrated together (identified via dependency analysis), so you move tightly coupled workloads in one cutover rather than breaking them apart.

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